

★ PRIMER · UNIT 4

Before the Lecture: The Straight Line

Unit 4 · Linear Models and Regularization

Watch this first if slope, intercept and least squares are hazy

Twenty minutes of groundwork — what a best-fit line is, what squared error measures, and what a penalty term does to a model that has too many clues.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Slope and squared error are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 4 Sits

The course builds one pipeline; here is this unit's stone

Unit 3

done

Labeling and meta-labeling.

Unit 4

you are here

The straight line, and how to
tame it.

Unit 5

next

Three classifiers and anomalies.

Every model after this is a straight line with something added.

What 'Best Fit' Means

A best-fit line is the one line that makes the total squared miss as small as possible.

The line

- Intercept — where it starts.
- Slope — how steep it is.
- Two numbers, one line.

The miss

- Each point sits off the line.
- Square each gap, add them.
- Least squares minimises that.

Squaring is why one wild point can drag the whole line towards it.

The Words the Lecture Assumes

Five terms, and the regularization section opens up

Word	What it means	Here
Intercept	the value when every input is zero	the baseline return
Slope	how much y moves per unit of x	the beta
Residual	actual minus predicted	what the line missed
Coefficient	the number multiplying a feature	one factor's weight
Penalty	a cost added for large coefficients	what tames the zoo

In finance a slope is called a beta. It is the same number under another name.

Three Penalties, Three Behaviours

The lecture compares all three; the difference is what they do to small coefficients

Method	Penalty on	What it does
OLS	nothing	fits freely, overfits happily
Ridge	squared coefficients	shrinks all of them, keeps all
Lasso	absolute coefficients	drives some to exactly zero
Elastic Net	both	shrinks, and still selects

Lasso selects; ridge only shrinks. That single difference decides which you reach for.

Why the Humble Line Often Wins

When the signal is faint and the data non-stationary, a simple model has less room to be wrong.

A complex model

- Fits every wrinkle of the past.
- Wrinkles do not repeat.
- Variance eats the gain.

A linear model

- Few parameters to get wrong.
- You can read every one.
- Often beats the fancy one.

In finance the burden of proof sits with complexity, not with simplicity.

A Four-Day Beta by Hand

Market return x , stock return y — compute the slope yourself

Day	x (market)	y (stock)	$x - \bar{x}$	$y - \bar{y}$
Mon	1.0	2.0	-1.0	-2.0
Tue	2.0	3.5	0.0	-0.5
Wed	2.0	4.5	0.0	+0.5
Thu	3.0	6.0	+1.0	+2.0
mean	2.0	4.0	—	—

Slope = $\Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2 = 4.0 / 2.0 = 2.0$. The stock moves twice the market.

Reading That Beta

One number, and two very different questions it answers

As a description

- Market up 1%, stock up 2%.
- It measures co-movement.
- Nothing causal is claimed.

As a risk number

- Twice the market's swing.
- Twice the drawdown too.
- Unit 11 prices this.

A beta describes how two things moved together — never why

The Factor Zoo Problem

Four hundred published factors, and what that does to a regression

With few clues

- Each coefficient is stable.
- You can interpret them.
- Out-of-sample holds up.

With four hundred

- Coefficients explode and flip.
- Signs reverse year to year.
- This is overfitting, again.

Unstable, sign-flipping coefficients are the tell. The lecture shows you the picture.

How Regularization Tames It

Three moves, and the lecture spends a slide on each

1 · Add a cost

for being large

Big coefficients now hurt the objective, not just the misses.

2 · Turn the dial

λ

More penalty, smaller coefficients, simpler model.

3 · Choose by test

not by taste

Cross-validation picks λ , and unit 10 makes that honest.

λ is not estimated from the data — it is chosen by how the model behaves out of sample.

Bias and Variance, in This Unit's Words

The same trade as unit 1, now with a dial you can actually turn

Lambda	Coefficients	Bias	Variance
zero	free to explode	low	very high
moderate	shrunk sensibly	some	much lower
very large	near zero	high	near zero

You are buying variance reduction with bias. The dial sets the exchange rate.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
β	beta	a coefficient — one feature's weight
ε	epsilon	the residual the line could not explain
λ	lambda	how hard the penalty bites
$\hat{y}$	y-hat	the model's prediction

Beta is finance's word for slope. Lambda is the only new dial in this unit.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>LinearRegression().fit(X, y)</code>	fit an ordinary least-squares line
<code>model.coef_</code>	the betas it found
<code>Ridge(alpha=1.0)</code>	the same, with a squared-coefficient penalty
<code>Lasso(alpha=0.1)</code>	the penalty that sets some betas to zero
<code>cross_val_score(m, X, y)</code>	score it on data it did not fit

In scikit-learn the penalty dial is called alpha, not lambda. Same thing.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 5

1

"A higher R-squared means a better model"

R-squared always rises when you add a feature — including a column of random numbers.

2

"Linear models are too simple for finance"

They win more often than they should, precisely because there is so little signal to fit.

3

"Ridge and lasso do the same thing"

Ridge shrinks every coefficient. Lasso sets some to exactly zero — that is feature selection.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 $\Sigma(x-\bar{x})(y-\bar{y})=4$, $\Sigma(x-\bar{x})^2=2$. What is the slope?

2.0

2 Which penalty can set a coefficient to exactly zero?

lasso

3 What happens to R-squared when you add a random column?

it rises

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It derives the line properly

Least squares, from the picture to the formula, with the four-day beta worked in full.

2

It opens the factor zoo

Four hundred published factors, and what happens to a regression that is fed all of them.

3

It ends in code

The problem set fits OLS, then ridge, and watches the coefficients stop flip-flopping.

Next: Unit 4 lecture — Linear Models and Regularization